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A Multi-Agent Generative AI Framework for Supporting Beyond Standard Model Physics Research



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Abstract

Recent advances in retrieval-augmented generation (RAG) and multi-agent language model systems have opened new opportunities for automating scientific document understanding. The complexity of high-energy physics literature, particularly Compact Muon Solenoid (CMS) Supersymmetry (SUSY) papers, presents significant challenges for automated interpretation and reasoning. This study introduces a multi-agent framework aimed at improving reasoning over highly technical physics documents and enabling efficient retrieval of textual content, tables, and figures. To achieve this, database construction workflows guided by dedicated agents are developed, resulting in enriched and retrieval-friendly representations of physics content. The system includes agents specialized in terminology interpretation, summarization, control evaluation, optimized description generation, and dedicated table and graph retrieval-and-reasoning. Textual content and tables are saved in enriched JSON formats and indexed into a vector database, while figures are stored in a structured SQL-based graph database. During retrieval, the Terminologist, Table, and Graph Agents identify relevant documents and pass them to the Question Answering Agent, which synthesizes the final response. The framework is evaluated through multiple experiments involving quantitative graph queries, conceptual physics questions, and table-based reasoning tasks. Results demonstrate that the system produces accurate, interpretable answers, improving accessibility to CMS SUSY literature and supporting automated scientific analysis.

1 Introduction

The advancement of technology and progress in physics research have been deeply intertwined. Fundamental research in physics has historically driven technological innovation, which in turn has accelerated the pace of discovery by enabling faster data acquisition, processing, and analysis. This relationship is perhaps most clearly demonstrated in the experiments at the Large Hadron Collider (LHC), particularly in the ATLAS and CMS experiments [1, 2]. Each experiment is equipped with its own dedicated detector, designed to record proton-proton collisions at an extraordinary rate—producing vast volumes of data per second. To conduct these experiments with greater precision and efficiency, continuous improvements in both detector technology and computational tools have been essential. While detector development focuses on improving sensitivity, resolution, and readout speed, advances in computing increasingly emphasize data processing, simulation, and, more recently, machine learning-driven analysis.

Machine learning has been employed in high-energy physics since its early use in charged particle tracking with neural networks [3], and it played a crucial role in the discovery of the Higgs boson by enhancing the sensitivity to its decay into τ leptons using boosted decision trees [4], illustrating its growing importance in LHC analyses [5]. As computing capabilities continued to advance, more sophisticated methods such as deep learning became viable, offering significant advantages for large and complex datasets. These developments have directly benefited high-energy physics experiments. Deep learning techniques have demonstrated strong performance in tasks such as particle identification, jet substructure analysis, and event classification, particularly in data-intensive scenarios [6].

Further advancements in deep learning have led to the development of large language models (LLMs), which have had a revolutionary impact across industry, academia, and a wide range of disciplines. These state-of-the-art AI systems are capable of understanding and generating human language. While earlier AI developments primarily focused on data-driven tasks such as classification or regression, LLMs introduce new capabilities including scientific reasoning, summarization, interactive document querying, idea generation support, and fine-tuned task adaptation—broadening their potential to assist in complex research workflows.

However, general-purpose LLMs often fall short when applied to physics, where symbolic reasoning, domain-specific terminology, and methodological rigor demand models that are specialized and aligned with disciplinary standards [7]. To address this challenge, new advancements and more sophisticated model architectures are required. One such approach is RAG, a framework that enhances the reasoning and factual grounding capabilities of language models by retrieving relevant external information and incorporating it at inference time [8].

For more complex architectures, multiple large language models can be specialized for particular tasks through techniques such as task-specific prompting, connecting to dedicated databases, or applying tailored reasoning strategies. These specialized models are often referred to as agents. By coordinating the behavior of such agents, multi-agent systems can be constructed that outperform single-model setups, enabling more robust, modular, and context-aware reasoning pipelines.

In this study, a multi-agent framework is developed to reason over physical context,

interpret domain-specific terminology, read and analyze tables and figures containing numerical data, link information to relevant textual content, and generate meaningful answers to user queries. To achieve this, distinct agents are designed to handle reasoning, database construction, and retrieval tasks. More specifically, the framework is tailored for high-energy physics and is applied to SUSY publications from the CMS Collaboration. The goal is to build a system that is aware of the literature produced by CMS in the context of Beyond the Standard Model (BSM) searches, and that can interpret, analyze, and answer questions related to the current state of research in this domain.

2 Related Work

Recent years have witnessed a surge of interest in using LLMs and agent-based AI architectures to automate and accelerate scientific discovery. In this section, we review key developments in multi-agent LLM frameworks, RAG, document understanding, and AI-powered scientific analysis, especially within complex technical domains such as high-energy physics.

2.1 Multi-Agent LLM Systems for Scientific Discovery

Multi-agent LLM frameworks have emerged as powerful tools for decomposing complex scientific reasoning into collaborative workflows. Google’s AI co-scientist system demonstrates how specialized LLM-based agents can autonomously generate, debate, and refine hypotheses, expediting cycles of scientific discovery [9]. Lu et al. [10] introduced the “AI Scientist,” an autonomous multi-agent system capable of not only hypothesis generation and experiment planning, but also manuscript writing and simulated peer review—completing an end-to-end research pipeline without human intervention. Recent work by Xu et al. [11] explored large-scale LLM discussion and review, showing that multi-agent debate can surpass single-model approaches in both scientific creativity and critical assessment.

2.2 Retrieval-Augmented Generation for Scientific Documents

RAG techniques have significantly improved LLM performance on scientific document tasks by grounding model outputs in relevant literature. Lála et al. [12] developed PaperQA, a RAG-based agent that answers complex questions over full-text scientific articles by retrieving and synthesizing supporting evidence, achieving near-expert performance on challenging benchmarks. RAG pipelines have also been employed for multi-modal and domain-specific retrieval, including systems that index tables, figures, and supplementary materials for enhanced factuality and transparency [8, 7].

2.3 Document Understanding and Multi-Modal Scientific QA

Automated understanding of scientific documents remains challenging due to their multi-modal nature—rich with equations, tables, and figures. The SPIQA dataset [13] and SciTabQA [14] enable benchmarking of multi-modal question answering (QA) in scientific papers, requiring models to jointly interpret textual and tabular/graphical content. Despite advances in visual-language models, recent studies report that QA over scientific tables and figures remains a difficult problem, with leading systems struggling on tasks involving quantitative comparison, unit conversion, or nuanced reasoning.

2.4 Domain-Specific LLMs for Physics and High-Energy Science

Domain-adapted language models such as Astro-HEP-BERT [15], AstroLLaMA [16], and PhysBERT [17] have been developed to address the vocabulary, conventions, and symbolic complexity of physics and astronomy literature. These models, pretrained on millions of scientific papers, outperform generic LLMs in document retrieval, entity linking, and concept disambiguation in their respective fields. However, even domain-specific models often lack robust capabilities for reasoning across text, tables, and figures, especially in highly technical documents such as those produced by the CMS collaboration [2].

2.5 Automated Table and Figure Extraction

Recent work has focused on extracting structured information from scientific tables and figures using a combination of vision-language models and LLMs. Methods like PlotExtract [18] employ chain-of-thought prompting and visual reasoning to digitize data from published plots, while tools such as ChatExtract leverage LLMs for accurate information extraction from text and tables. These developments are enabling large-scale automated curation of scientific data from the literature.

2.6 Positioning of This Work

While prior systems have demonstrated the power of multi-agent LLMs, RAG pipelines, and domain adaptation for scientific literature, most existing approaches either focus on biomedical domains or lack the modular, agent-oriented design necessary for interpretable, step-by-step reasoning in physics. Our framework uniquely combines terminology interpretation, table and graph extraction, multi-modal retrieval, and collaborative reasoning via specialized agents—tailored specifically to the complexity of CMS SUSY literature. By integrating these components into a unified, extensible architecture, our system advances the state of the art in explainable scientific QA and automated literature analysis for high-energy physics.

3 Physics Beyond the Standard Model

The CMS is one of the major particle detectors operating at the LHC at CERN. It is designed to study high-energy proton-proton collisions and to investigate fundamental questions in particle physics, including the origin of mass, the nature of dark matter, and the search for physics BSM.

CMS publications provide detailed, peer-reviewed analyses that systematically document the results of experimental searches for BSM phenomena. These papers contain a wealth of structured scientific information, including theoretical motivations, precise measurements, exclusion limits, and interpretations of possible new physics scenarios. Their rigorous format and standardized structure make them an ideal corpus for developing automated tools for scientific analysis.

In this study, CMS SUSY papers are selected as the primary source for developing and testing an AI assistant capable of interpreting and reasoning over complex physics content. The system is designed to answer scientific questions, analyze figures and tables, compare results across different papers, and support future research directions in BSM physics through the use of large language models and structured retrieval mechanisms.

4 System Architecture

This section presents the architecture of the proposed system, including the definition of individual agents, the structure of the underlying databases, and the overall workflow. CMS SUSY papers contain high-level scientific content, often involving domain-specific terminology, dense symbolic expressions, and complex figures that can be difficult to interpret—even for PhD-level physicists. As such, enabling LLMs to reason over and extract meaningful information from these documents requires a carefully designed multi-agent framework.

To address these challenges, the system defines a set of specialized agents, each responsible for handling a specific bottleneck in the interpretation and retrieval of physics knowledge. This section first outlines the main challenges in understanding CMS SUSY papers, then describes the agents designed to overcome these challenges, and finally explains the structure of the databases created using their outputs to enable efficient retrieval and reasoning.

All agents in the system are implemented using the Gemini 2.0 Flash and Gemini 2.5 Flash large language models [19], selected for their fast inference and strong performance on scientific reasoning tasks.

4.1 Challenges

CMS SUSY papers contain advanced theoretical and experimental content that requires a strong background in both physics and mathematics to interpret correctly. Many of the terms and concepts presented in these papers are highly specialized and often require deeper, more fundamental explanations to provide meaningful insight. This presents a challenge for LLMs, which are not inherently equipped to understand domain-specific context without additional guidance.

Moreover, these papers rely heavily on symbolic expressions and mathematical notation. Since the documents are processed from LaTeX source files, many physics terms appear as LaTeX-style abbreviations or equations—formats that are difficult for LLMs to interpret directly without prior conversion into natural language.

Another significant challenge lies in the structured content such as tables and figures. While standard methods exist for parsing and chunking plain text, the semantic interpretation of tables and graphs remains difficult. Reasoning over such content and linking it back to the surrounding narrative is essential for a comprehensive understanding of the analysis, but is non-trivial to automate.

Beyond isolated interpretation, many scientific questions refer to multiple modalities—for example, asking about the relationship between a figure and a table, or between textual context and numerical data. Answering such questions requires coordinated, cross-modal reasoning that standard single-agent systems struggle to handle.

Additionally, these papers are often long and complex, making retrieval a non-trivial task. Efficient access to relevant figures, tables, and text sections demands a high-quality retrieval strategy that balances semantic understanding with keyword-based precision. This motivates the use of hybrid methods such as dense vector search, re-ranking, and metadata-aware indexing.

Finally, scientific content in these papers often lacks explicit grounding. Figures may be referenced without clear contextual explanation, or terms may appear without prior definition. LLMs must therefore be supplied with structured context—such as section summaries and standardized descriptions—to reason effectively over such documents.

4.2 Agent Definitions

To address the challenges outlined in the previous subsection, a set of specialized LLM agents is defined. Each agent is responsible for a specific task in the system, contributing to reasoning, database construction, semantic enrichment, and final answer generation. These agents work both independently and collaboratively to ensure modularity and interpretability within the pipeline.

Terminologist Agent: This agent is responsible for translating LaTeX-style physics terms and symbolic equations into their natural language counterparts. For example, expressions such as $\vec{p}_T$ are mapped to "transverse momentum." Beyond translation, it provides concise explanations and contextual use cases for each term, leveraging the surrounding section text. This ensures that other agents—particularly those involved in table and figure reasoning—can interpret and reuse these terms consistently across different modalities. The Terminologist Agent thus enables the system to operate with a shared scientific vocabulary.

Table Agent: This agent is a hybrid retrieval-and-reasoning component designed to answer scientific questions related to tables in CMS SUSY papers. It combines semantic and keyword-based search over table captions, headers, and enriched cell contents—augmented with technical explanations from the Terminologist Agent—to retrieve relevant tables and associated context. It then performs structured querying over the JSON-formatted table data to extract both descriptive and quantitative information needed to answer user queries accurately.

Graph Agent: This agent is a hybrid retrieval-and-reasoning component designed to answer scientific questions about figures in CMS SUSY papers. It performs semantic similarity search over figure metadata using vector embeddings and complements this with SQL-based numerical querying over structured bin data stored in a relational database. By combining textual and quantitative information, it retrieves relevant figures, interprets their content, and extracts precise numerical values needed for analysis and reasoning.

Question/Answer Agent: This agent is an advanced reasoning and synthesis module that integrates outputs from multiple agents—including the Table Agent, Terminologist Agent, and Summarizer Agent—to generate comprehensive, context-aware responses to user queries. It combines structured data from tables, natural language explanations of technical terms, and relevant textual content to construct detailed and scientifically grounded answers. Designed to handle both simple factual questions and complex analytical queries, it coordinates reasoning across modalities to deliver accurate and interpretable results.

Summarizer Agent: This agent generates concise summaries for each section and for the entire paper. These summaries serve as contextual scaffolding for downstream agents—particularly those handling figure and table retrieval—by providing a high-level understanding of the paper’s structure and content. By capturing the key ideas and scientific goals of each section, it helps other components ground their reasoning in the

appropriate context, improving both accuracy and relevance during answer generation.

Description Agent: This agent produces optimized, search-friendly descriptions for figures and tables by leveraging axis labels, metadata, and section-level summaries. Its goal is to generate semantically rich and compact text representations that improve retrieval quality during both vector-based and keyword-based searches. These enhanced descriptions are embedded and stored alongside other metadata, allowing the system to retrieve relevant visual content more accurately in response to user queries.

Re-Ranking Agent: This agent is used during the graph retrieval process to improve selection accuracy. After candidate figures are retrieved via semantic and keyword-based search, it evaluates their relevance to the user’s question using LLM-based semantic reasoning. It then selects the best-matching figure by reordering the candidates based on contextual alignment.

Control Mechanism Agent: This agent plays a quality control role within the graph retrieval pipeline. It evaluates the generated answer alongside the retrieved figure to determine how well they address the user’s question. Based on this evaluation, it returns a verdict—either “GOOD” or “BAD”—and may optionally provide feedback or suggestions for improvement. This step ensures that answers are both scientifically accurate and contextually relevant before being presented to the user.

4.3 Database Construction and Indexing

Database construction is a critical component of the system, as it enables structured reasoning, efficient retrieval, and ultimately more accurate answers. Much of the reasoning performed by the agents depends on how well the underlying data is parsed, enriched, and stored. Therefore, careful design and consistent formatting are essential.

CMS SUSY papers primarily contain three types of data: textual content, tables, and figures. These components are naturally separated in the LaTeX source code, making them feasible to parse. For each paper, the document is processed and converted into a structured JSON file containing enriched sections, extracted tables, and references to figures.

While textual content and tables are directly available in the LaTeX source, figures are typically included as image files and lack embedded numerical data. To address this limitation, figure metadata and bin-level numerical values are obtained from HEPData, a curated repository of experimental results from the CMS and ATLAS collaborations. Currently, figure data is available for approximately 21.6% of CMS SUSY papers. This HEPData integration is essential for enabling the system to answer questions involving plots—such as event yields, exclusion limits, or cross-section comparisons.

The structured outputs—section texts, term explanations, figure metadata, bin data, and table contents—are stored in two main database systems: a relational SQL database and a semantic vector index, as shown in Figure 1. Each database construction process is handled separately.

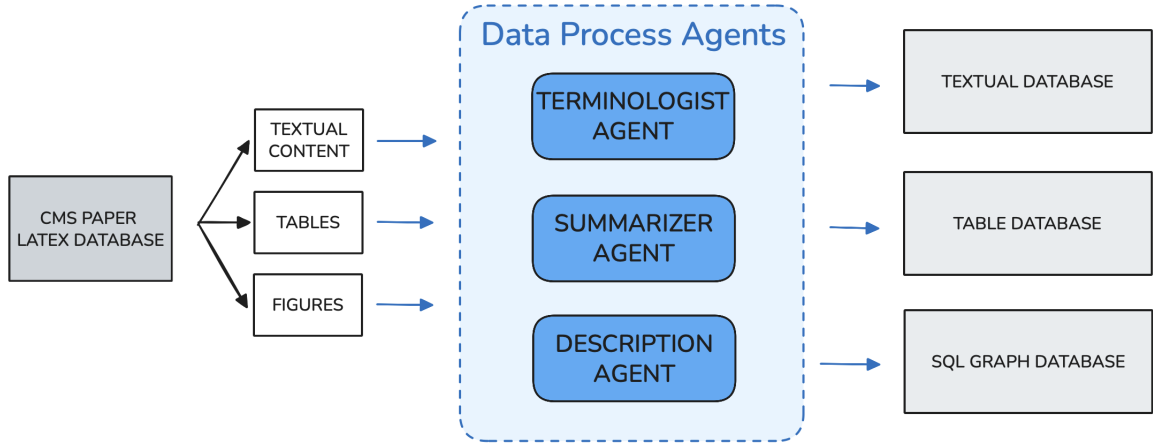


Figure 1: General Database Construction Schematic

Although the pipelines differ, the defined agents are used throughout each process. Each pipeline is described in more detail below.

Textual Database: This pipeline is primarily supported by the Terminologist Agent, which plays a central role in enriching the scientific language of CMS SUSY papers. After parsing each LaTeX document, the textual content is divided section by section and stored in a structured JSON file. To enable content-aware reasoning, each LaTeX-style physics term—whether symbolic or command-based—is individually sent to the Terminologist Agent along with the section text. The agent generates a natural language equivalent, along with a brief explanation, contextual use case, and pronunciation.

Once all terms are interpreted, they are replaced within the section text by their natural forms and annotated with explanations reflecting their contextual meaning. This enriched content is saved alongside the raw and indexed text within the same JSON structure, as illustrated in Figure 2.

The processed files are then embedded using the `allenai-specter` model and stored in a FAISS index for semantic retrieval. The resulting vector store—referred to as the Textual Database—allows user queries to be matched against the most relevant sections, improving the grounding and interpretability of answers.

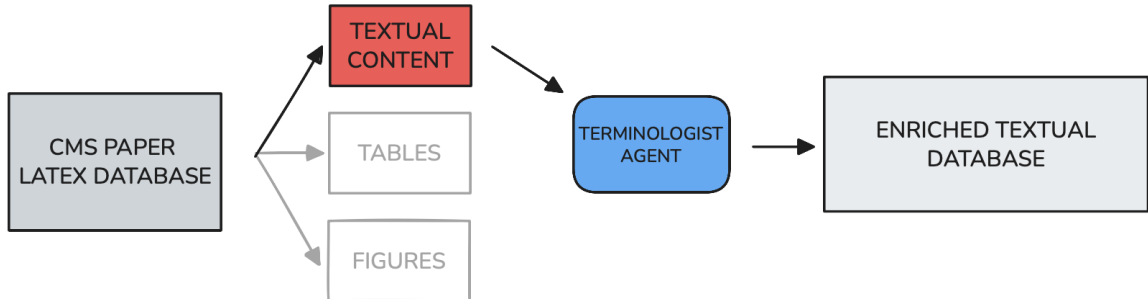


Figure 2: Textual Database Construction Schematic

Table Database: In contrast to figures or narrative sections, the tabular content in CMS SUSY LaTeX papers provides rich structured numerical information, yet remains

challenging for automated parsing and reasoning due to LaTeX-specific syntax, abbreviations, and lack of context. To address this, a dedicated Table Database is constructed by parsing, enriching, and organizing all table blocks into a structured format that enables both semantic and numerical retrieval.

All table blocks are first extracted from the LaTeX source using regular expressions targeting `\begin{table}...\end{table}` environments. Each block is then parsed to isolate metadata such as captions and labels, along with tabular content including headers and row values. These components are normalized into a JSON schema that captures both structural and semantic information.

To embed scientific context, surrounding paragraph-level text and section titles are linked to each table, forming a `context` field. This enables the system to understand not just what the table conveys, but in which part of the scientific argument it appears.

Next, all table cells are scanned for LaTeX-style physics terms and symbolic expressions. These are resolved using the Terminologist Agent, which provides natural language translations and contextual explanations for each identified symbol (e.g., p_T , $\bar{b}$). The enriched cells are stored as `columns_explanations`, making each table semantically self-contained and machine-interpretable.

Finally, all enriched tables are inserted into a structured table database in both JSON and SQL-compatible formats. The database supports hybrid querying: keyword and semantic search over captions and cell content, along with structured querying over numerical rows. This design enables scientific agents to retrieve relevant tables, perform comparisons across papers, and carry out reasoning involving uncertainties, systematics, or control-signal region differences.

In later stages, this structured database is also used to support cross-table analyses, such as detecting mismatches or validating consistency in results across different signal regions or theoretical benchmark points. An example is illustrated in Section 5.2.

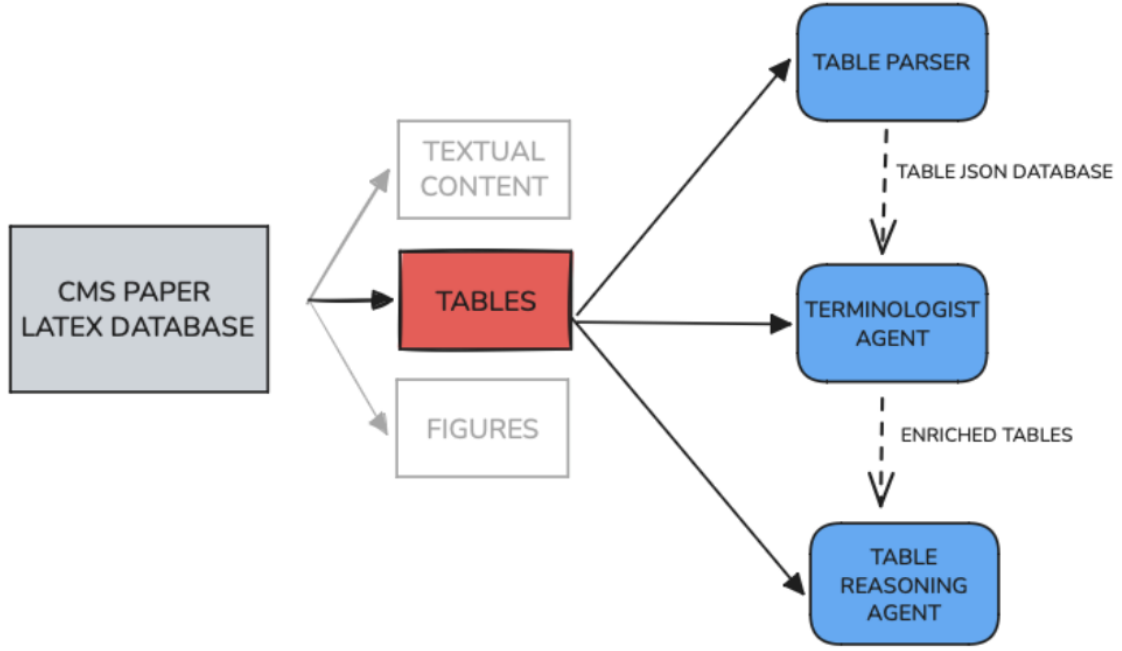


Figure 3: Table Database Construction Schematic

SQL Graph Database: Figures in CMS SUSY LaTeX documents are typically included as static image files, without accessible numerical data. To enable reasoning over figures, external data is required. For this purpose, HEPData is used as a complementary source, providing numerical bin values and associated metadata for a subset of CMS figures. This information is essential for answering scientific questions involving histograms, exclusion plots, and cross-section curves.

Once retrieved from HEPData, figure metadata is semantically enriched using the Terminologist Agent, which translates LaTeX-style physics terms into natural language. The Summarizer Agent provides contextual grounding through section and paper summaries, while the Description Agent generates optimized, search-friendly captions based on axis labels and metadata. All enriched textual content is embedded using the `allenai-specter` model and stored via `pgvector` for semantic retrieval. Simultaneously, figure-level metadata and numerical bin data are inserted into a structured SQL schema consisting of `hep_figures` and `hep_bins`, as illustrated in Figure 4. This setup enables both semantic figure retrieval and precise SQL-based numerical queries.

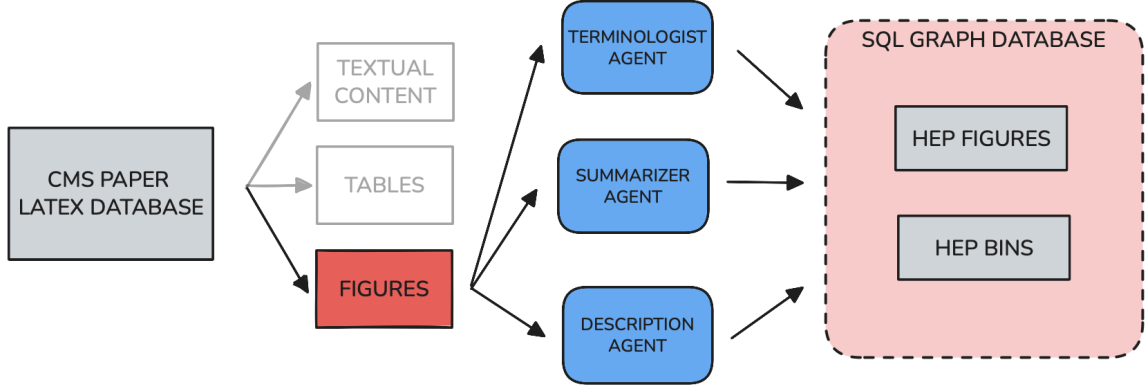


Figure 4: SQL Graph Database Construction Schematic

With this comprehensive database construction pipeline in place, the system is now equipped to support robust retrieval methodologies and enable accurate, context-aware responses to user questions.

5 Retrieval Methodologies and Final Answer

Document retrieval is one of the most critical components of the proposed framework. All preceding steps—including parsing, term interpretation, summarization, and database construction—are designed to enhance the quality and relevance of the documents retrieved in response to user queries. The goal is to ensure that the system not only identifies the correct information but also interprets its scientific context accurately.

This section first describes the retrieval pipelines for the three primary sources of information: the textual database, the table database, and the SQL-based graph database. For each, the semantic and keyword-based retrieval strategies are explained, along with how enriched metadata and embeddings improve performance. Finally, the role of the Question/Answer Agent is introduced, detailing how it integrates information retrieved from different sources and synthesizes a coherent, context-aware answer tailored to the user’s query. An overview of this multi-stage retrieval and reasoning process is illustrated in Figure 5.

5.1 Textual Retrieval Pipeline

A key step in constructing the textual database was the semantic enrichment of each section using the Terminologist Agent. This agent replaced LaTeX-style physics terms with their natural language equivalents and added brief explanations based on the surrounding section context. This enrichment significantly enhanced the semantic coherence of the text and improved retrieval performance.

Each section was then divided into smaller, manageable chunks and indexed using a FAISS vector store. Every chunk was labeled with a unique identifier encoding the paper ID and section ID, enabling precise tracking and retrieval. The embeddings were generated using the `allenai-specter` model, which is specifically designed for scientific text.

During retrieval, a user’s question is embedded using the same model and compared to the stored chunk embeddings. The system retrieves the top three semantically relevant

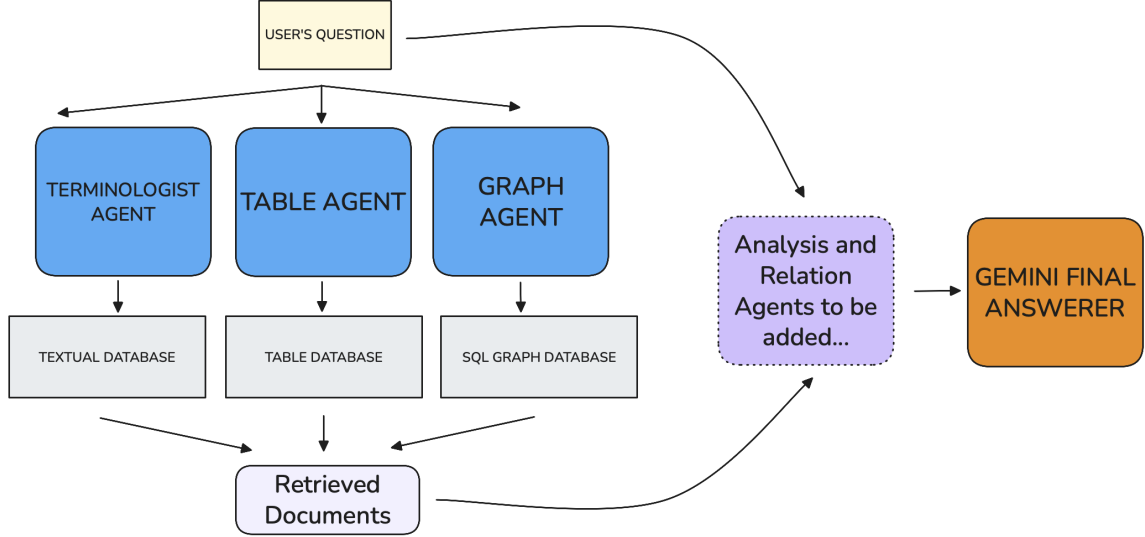


Figure 5: General Document Retrieval and Question-Answer Pipeline

text segments from the vector index. It is important to note that this textual retrieval pipeline operates independently from the table and graph retrieval pipelines, allowing specialized optimization for natural language reasoning.

5.2 Table Retrieval Pipeline

The table retrieval pipeline is designed to enable scientific question answering over structured numerical data extracted from CMS SUSY papers. This process begins with the semantic indexing of each table, enriched through multiple stages of preprocessing and annotation.

First, all table captions, headers, and cell contents are processed and normalized. Using the **Terminologist Agent**, LaTeX-style physics terms in the cells are mapped to their natural language equivalents. Each enriched cell is stored alongside its explanation in a structured format, allowing agents to interpret table content with domain-specific understanding.

Next, the surrounding context—including section titles and adjacent paragraphs—is associated with each table. This contextual information enables reasoning not only over raw data but also over the scientific purpose of each table (e.g., signal yield comparison, systematic uncertainties, background predictions).

Each table, represented in enriched JSON format, is then embedded semantically using a dual indexing strategy. Captions and textual fields are vectorized using the **allenai-specter** model and indexed in a **FAISS**-based vector store, while keyword fields (such as row labels and term explanations) are indexed using a sparse **BM25** engine in PostgreSQL full-text search. This hybrid retrieval strategy ensures both semantic sensitivity and keyword precision.

When a user submits a question, the Table Agent identifies whether the intent involves lookup, comparison, statistical inference, or summary. Based on this, the agent queries both the vector and keyword indices to retrieve the top relevant tables. These are then passed to a reasoning module that:

- isolates relevant rows or columns,
- cross-references technical terms using the Terminologist Agent,
- performs numeric analysis if necessary (e.g., uncertainty propagation or normalization checks),
- and generates a concise, referenced natural language answer.

In addition to single-table analysis, the Table Agent supports cross-table reasoning. For this, a row- and column-level **cosine similarity matrix** is computed across numerical values and headers of different tables. This matrix is used to align semantically or statistically similar entries, enabling the comparison of data from different papers or experimental setups. As a result, values such as signal yields, background estimates, or systematic errors can be matched and jointly interpreted—even when they originate from structurally different tables.

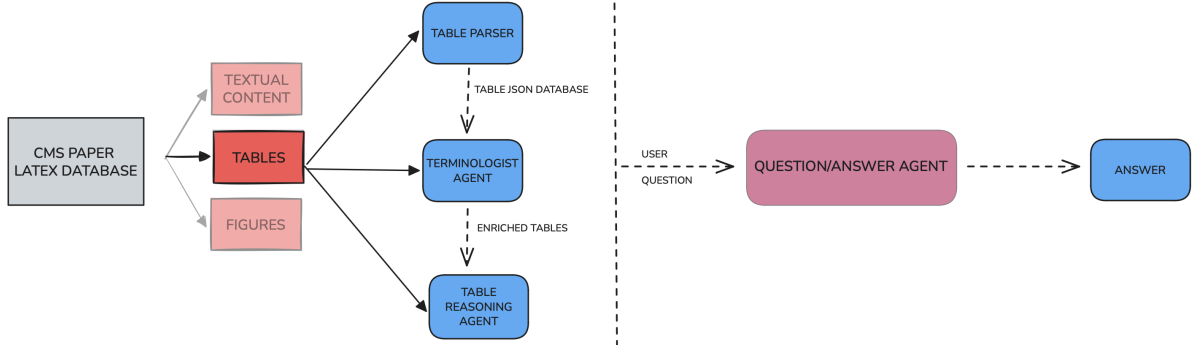


Figure 6: Overview of the table retrieval and reasoning pipeline. Tables are parsed, semantically enriched, and passed through a multi-agent system that ultimately routes user questions to data-driven answers.

This modular pipeline allows the Table Agent to answer complex physics questions that require a combination of table lookup, symbol interpretation, and scientific reasoning—independently of figure or textual data.

5.3 Graph Retrieval Pipeline

The detailed construction of the SQL Graph Database is described in Section 4.3. In this subsection, the focus shifts to the retrieval process—specifically, how relevant figures are identified and extracted from the database in response to user queries.

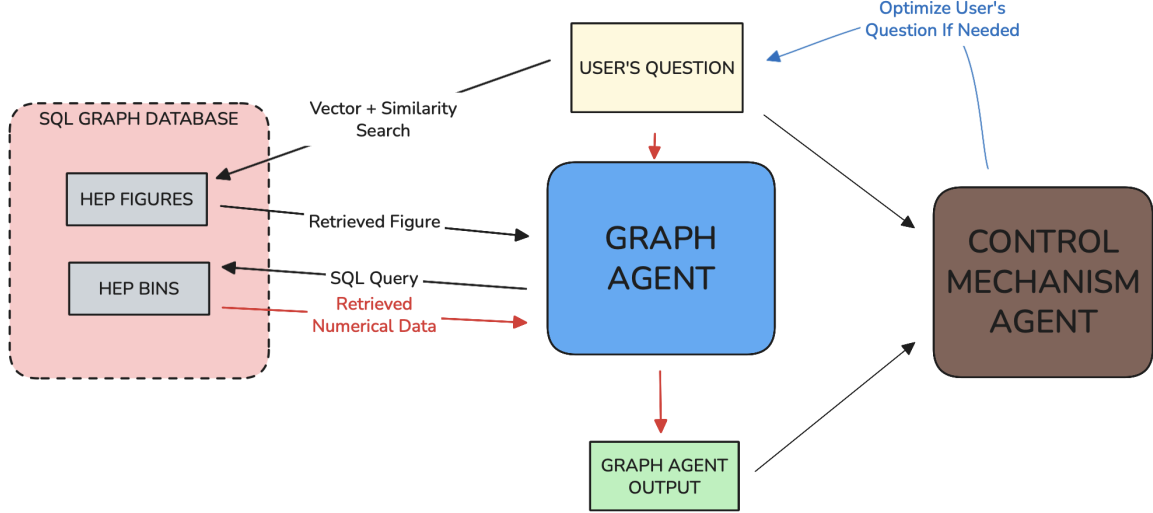


Figure 7: Figure Retrieval Pipeline

As illustrated in Figure 7, once a question is submitted by the user, the Graph Agent initiates a hybrid retrieval process to identify the most relevant figures from the SQL graph database. This involves a combination of semantic vector search—using `pgvector` and `allennai-specter` embeddings—and sparse keyword-based search via PostgreSQL’s full-text search functionality. The results from these two modalities are merged using Reciprocal Rank Fusion (RRF), which balances semantic similarity and keyword overlap to generate a robust set of candidate figures.

To enhance the precision of figure selection, a semantic re-ranking stage is applied using the Gemini model. In this stage, each candidate figure—retrieved through hybrid search—is evaluated in the context of the user’s question and its associated metadata. The objective is to identify the figure that most accurately aligns with the intent and semantic content of the query. In cases where the selected figure contains numerical information, such as event yields or exclusion limits, the system invokes Gemini to construct a context-aware and syntactically correct SQL `SELECT` query. This query is used to retrieve the relevant bin-level data from the `hep_bins` table, enabling fine-grained quantitative reasoning.

The final answer is composed based on outputs retrieved from various agents and databases and is delegated to the Question/Answer Agent for synthesis. After the answer is generated, it is passed to the Control Mechanism Agent for evaluation. This agent assesses whether the answer is relevant to the user’s question and whether the correct figure has been retrieved. It then returns a verdict of either `GOOD` or `BAD`. In cases where the result is marked `BAD`, the Control Mechanism Agent provides feedback and uses the original question to generate an enhanced version of the query. The retrieval and reasoning pipeline is then re-executed using this refined query to improve the quality and relevance of the final output.

5.4 Question-Answer Pipeline

The Question/Answer (Q&A) Agent serves as the central synthesis module in the multi-agent architecture, responsible for generating scientifically grounded responses by inte-

grating information from diverse modalities: structured tables, technical terminology, textual context, and optionally, graphical data. Unlike retrieval agents that operate on a single modality, the Q&A Agent orchestrates collaboration across the Table, Terminologist, and Text Agents, aligning their outputs to address user questions with precision and clarity.

The reasoning process begins when a user submits a question. The Q&A Agent first determines which modalities are required to interpret the query—assessing whether the question involves structured data (tables or figures), domain-specific terms, or textual context. Once identified, the agent dispatches specialized subqueries to the corresponding retrieval modules:

- The **Table Agent** extracts structured tables relevant to the question, performing data alignment and value matching.
- The **Terminologist Agent** identifies scientific terms and provides enriched, context-aware definitions.
- The **Text Agent** retrieves relevant narrative sections or summaries from surrounding paragraphs and section titles.
- Optionally, the **Graph Agent** queries SQL-based figure metadata to retrieve visual data when required.

All retrieved content—JSON-formatted tables, enriched technical terms, and contextual text—is then forwarded to the Q&A Agent. A collaborative reasoning routine is triggered, where the agent performs structured synthesis in the following sequence:

1. **Data Alignment:** Each data source is parsed and semantically aligned to ensure consistent interpretation across modalities.
2. **Agent-Based Subtasks:** Reasoning steps are delegated to sub-agents: quantitative inference to the Table Agent, definition mapping to the Terminologist Agent, and narrative structuring to the Text Agent.
3. **Cross-Modality Fusion:** Outputs from each sub-agent are aggregated into a unified knowledge representation, linking values to definitions and contextual roles.
4. **Final Answer Generation:** A response is synthesized using step-by-step reasoning, scientific clarity, and explicit references to the underlying data and concepts.

In JSON-based mode, the agent uses structured prompts to instruct each sub-agent. For instance, the Table Agent prompt includes table metadata, selected rows, and contextual notes; the Terminologist prompt supplies a list of relevant terms; and the collaborative prompt template merges these into a stepwise answer generation task. This prompt-based orchestration enables the Q&A Agent to leverage external language models (e.g., Gemini 2.0) for natural language synthesis while preserving scientific rigor.

In SQL mode, the Q&A Agent queries PostgreSQL databases using metadata about stored figures, schema definitions, and sample rows. Depending on the selected configuration, reasoning may involve graph-only, hybrid (graph + table), or full multi-modal pipelines that also incorporate domain-specific terminology.

Figure 8 illustrates the full multi-agent Q&A pipeline, highlighting how modular agents

collaborate to retrieve, interpret, and synthesize information for accurate, explainable scientific answers.

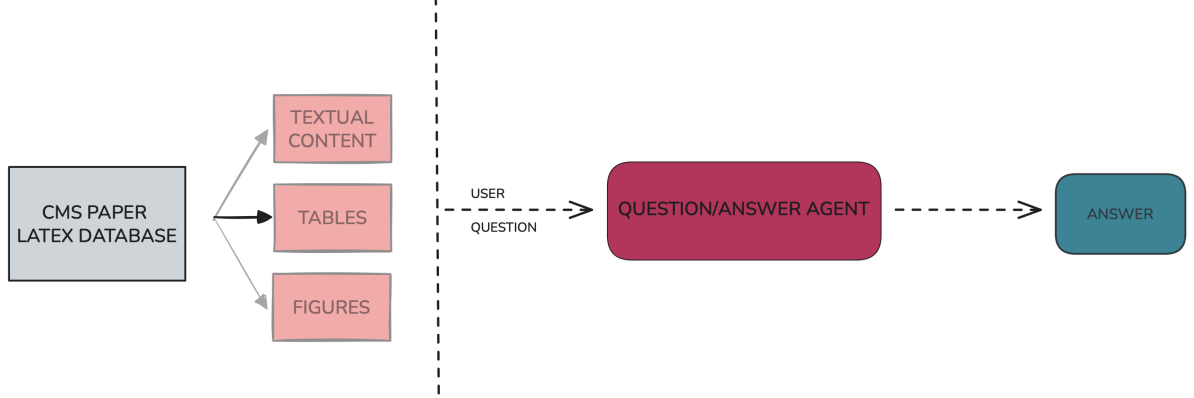


Figure 8: The Q&A pipeline integrates outputs from Table, Terminologist, Text, and optionally Graph agents. User questions are parsed, relevant modalities are queried, and a multi-step reasoning engine generates structured answers.

6 Results

This section presents a series of experiments designed to evaluate the effectiveness of the proposed multi-agent framework in retrieving and reasoning over scientific content from CMS SUSY papers. The performance of individual agents—including the Terminologist, Graph, and Table Agents—is demonstrated through selected use cases involving both descriptive and quantitative queries. Each example illustrates how the system interprets user questions, retrieves relevant information, and generates accurate, context-aware responses.

The evaluation is structured around two key criteria: (1) how effectively the system addresses the challenges identified earlier—such as symbolic notation, cross-modal reasoning, and scientific context interpretation—and (2) how useful and interpretable the generated answers are from a physicist’s perspective. CMS SUSY papers often contain highly technical content that can be difficult to interpret, even for PhD-level researchers. In particular, understanding figures, tables, and symbolic terms frequently requires significant time and expertise.

The proposed system aims to lower this barrier by providing high-quality, semantically grounded answers that assist physicists in navigating and interpreting the content more efficiently. By automating parts of the interpretation process and linking symbolic content with contextual explanations, the framework supports faster comprehension and more accessible engagement with complex scientific literature.

All performance evaluations in this study are conducted using a single CMS SUSY publication: *Search for new phenomena in final states with jets and missing transverse momentum in proton-proton collisions at $\sqrt{s} = 13$ TeV* [20]. The textual content, tables, figures, and question–answer pairs used in the experiments are all derived from this paper.

6.1 Terminologist Agent Evaluation

This experiment evaluates the system’s ability to respond to a conceptual question that requires structured scientific reasoning rather than numerical extraction. A question was posed asking for the difference between Control Regions (CRs) and Signal Regions (SRs), including how these regions are defined and used in high-energy physics analyses.

The generated response accurately identifies the role of CRs in constraining background predictions and the purpose of SRs in testing for new physics signals. It further explains how these regions are defined using observables and optimized selection criteria, emphasizes the importance of orthogonality between CRs and SRs, and describes the use of Validation Regions (VRs) to test modeling strategies. A realistic example is provided, and the answer concludes with a concise summary table comparing CRs and SRs across multiple dimensions.

Table 1 presents a checklist of key conceptual components and indicates their presence in the generated answer. All criteria are satisfied, demonstrating the system’s ability to deliver a complete and pedagogically structured explanation. The quality of the response can be attributed to the enriched section text, term expansions, and domain-aware embeddings generated during preprocessing. Overall, the system proves capable of addressing abstract scientific questions with clarity and precision, making it a useful tool for both researchers and students engaging with complex methodological material.

Table 1: Checklist for Conceptual Reasoning over SR and CR

Key Concept	Covered in Answer?
Definition of Signal Region (SR)	✓ Clearly defined with motivation to maximize sensitivity to signal using optimized cuts
Definition of Control Region (CR)	✓ Explained as regions used to constrain backgrounds, kinematically similar to SR
Step-by-step construction (cuts, variables)	✓ Observables, selection criteria, signal/background separation are described
Orthogonality of SR/CR	✓ Explained as crucial to minimize correlation
Role of Validation Regions (VRs)	✓ Included to check consistency of background estimation
Practical Example	✓ A diphoton search scenario is included
Tabular summary of differences	✓ A table comparing SR and CR is provided
Use of enriched and structured text	✓ Structure follows scientific reasoning with headings, lists, and domain terminology

6.2 Table Agent Evaluation

This experiment evaluates the framework’s ability to analyze tabular data and identify dominant sources of systematic uncertainty. A question was posed asking which systematic uncertainties, as listed in Figure 9, had the largest impact, and which affected both shape and normalization versus only normalization.

The answer generated by the model demonstrates correct interpretation of both the table’s

structure and semantics. It identifies the two largest shape-related uncertainties—*b tag efficiency* (8%) and *incorrectly b-tagged jets* (7%)—and explains that the uncertainty type “shape + lnN” refers to effects on both the distribution shape and the overall normalization. The response is well-organized into distinct steps: extraction, reasoning, synthesis, and final conclusion. It also correctly notes the table’s cutoff at 0.5% and that normalization-only uncertainties are not explicitly separated.

A notable strength of the system is its ability to reason about implicit constraints: it correctly infers that only uncertainties with an impact greater than 0.5% are listed in the table and discusses the implications of this threshold. Furthermore, the distinction between shape-affecting and normalization-only uncertainties is made based solely on the uncertainty type, without requiring explicit prompting.

While only a subset of the most impactful uncertainties is mentioned, the answer demonstrates a solid understanding of both numerical values and categorical labels used in the table. This indicates that the Table Agent can reason beyond surface-level content and synthesize information into scientifically meaningful insights.

Figure 9 illustrates the user query, the extracted table, and the generated system response.

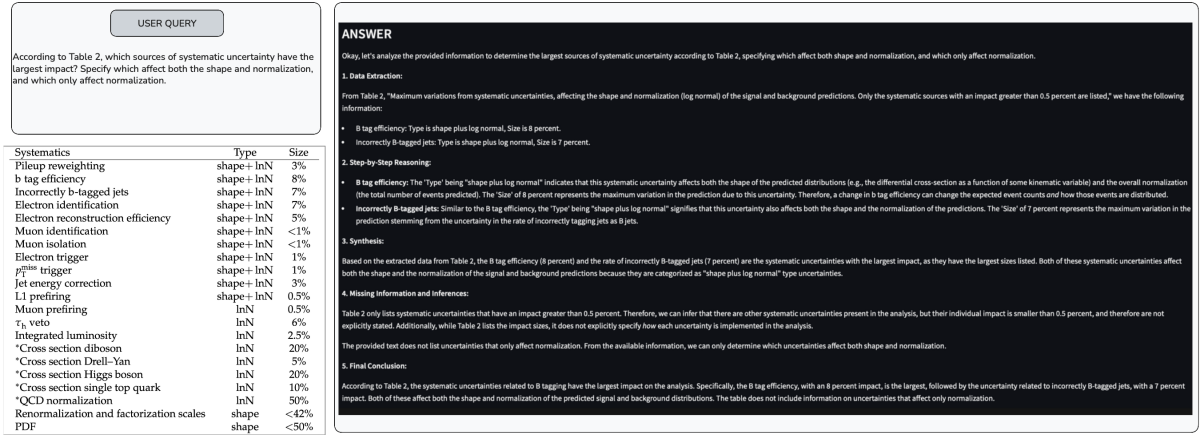


Figure 9: Table Experiment

6.3 Graph Agent Evaluation

Interpreting figures in CMS SUSY papers poses a significant challenge, as it requires not only technical expertise in high-energy physics but also a deep understanding of the specific analysis being conducted. Figures often present complex relationships related to new physics searches, statistical comparisons, or potential deviations from the Standard Model. Effective interpretation demands awareness of the analysis context, the physical meaning of the variables shown, the definitions of regions such as Signal Regions, and the quantitative details—including event yields, uncertainties, and axis labels.

To meet these demands, the proposed framework relies on the collaborative operation of multiple agents during both database construction and retrieval. When a user submits a question, the Graph Agent retrieves the most relevant figure, optionally queries numerical bin data, and passes the information to the Question/Answer Agent. This pipeline ensures that all necessary content is available for generating scientifically accurate and context-aware responses.

The following example demonstrates how the system interprets and reasons over figures in a CMS SUSY paper.

Experiment:

This experiment evaluates the system’s ability to respond to a query requiring reasoning over a figure from a CMS SUSY paper. A question was posed requesting a comparison between the observed and expected event yields in the high missing transverse momentum ($p_T^{\text{miss}} > 550$ GeV) region of the 1b Signal Region, as presented in Figure 10, along with its implication for the overall search result. The system successfully identified the relevant figure and extracted both the observed value (669.0 events) and the postfit prediction (647.28 events), including their respective uncertainties. It then evaluated the statistical significance of the difference and concluded—consistent with the Signal Region analysis—that the data showed no significant excess over the Standard Model prediction. The final answer also included an interpretation regarding upper limits on dark matter production.

All key elements from the evaluation checklist were present in the generated response, demonstrating the framework’s ability to perform both quantitative extraction and contextual reasoning. Such outputs may assist physicists by reducing the time required to interpret plots and draw conclusions, particularly when navigating complex multi-figure analyses.

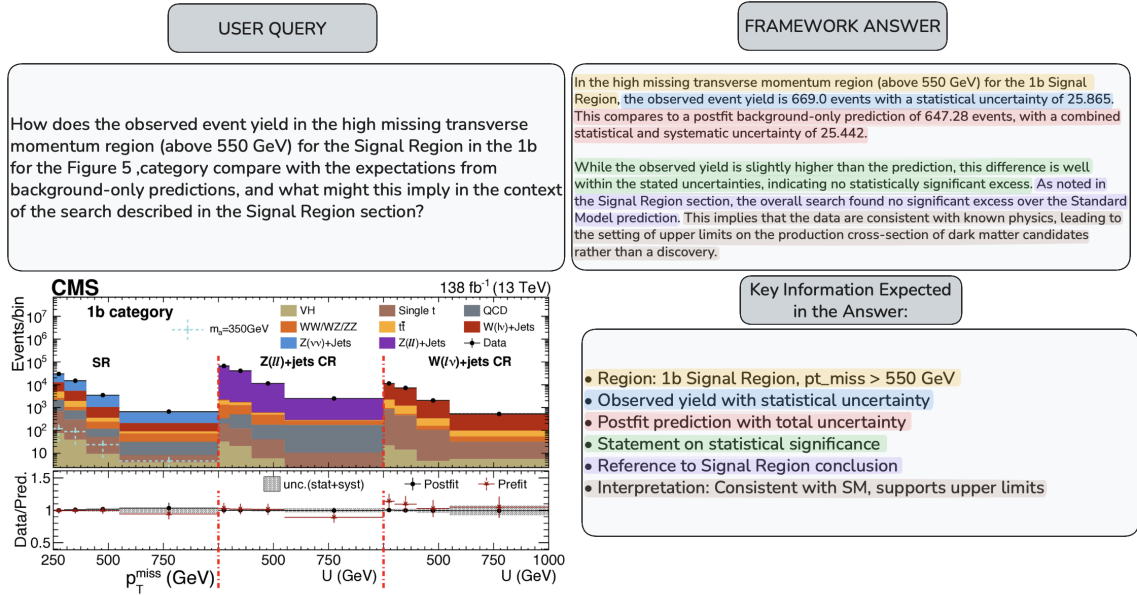


Figure 10: Figure Experiment

7 Conclusion

In this study, a multi-agent framework was developed that performs structured reasoning, database construction, and document retrieval over CMS SUSY papers. The system interprets complex scientific content—including symbolic terms, figures, and tables—by employing dedicated agents for term interpretation, table processing, and graph reasoning. The roles and processing pipelines of key components, such as the Terminologist Agent, Table Agent, and Graph Agent, were described in detail.

The framework’s effectiveness was demonstrated through a series of experiments, which showed that the system can retrieve relevant content, extract quantitative data, and generate accurate, context-aware answers to physics-related queries. These results highlight the potential of LLM-driven modular systems to support high-energy-physics research by reducing the time needed to interpret technical documentation and by improving access to analysis results.

Future work will focus on scaling the framework to a broader set of CMS SUSY publications, thereby enabling large-scale comparison, automated summarization, and the discovery of recurring patterns and insights across the literature.

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